

Credit Risk Measurement

Advanced Topics in Banking - Project 2

Group 4 | WU Wien, SS 2026

Caterina Piacentini · Farkas Tallos · Balazs Limpek

2026-05-24

Contents

1	Task	7
1.1		7
2	Task	8
2.1		8

```
knitr::opts_chunk$set(  
  echo = TRUE,  
  warning = FALSE,  
  message = FALSE,  
  fig.width = 7,  
  fig.height = 5,  
  fig.pos = "H",  
  out.width = "100%"  
)
```

```
if(!require("pacman")) install.packages("pacman")
```

```
## Loading required package: pacman
```

```
pacman::p_load(  
  data.table,  
  tidyverse,  
  knitr,  
  kableExtra,  
  ggplot2,  
  openxlsx,  
  dplyr  
)
```

```
BASE <- "/Users/farkastallos/Library/CloudStorage/OneDrive-WUWien/00_WU/01_2_YEAR/03_Q3/02_ATB/
```

```
ratings <- read.xlsx(paste0(BASE, "Data_Group_4.xlsx"), sheet = "Ratings")
portfolio <- read.xlsx(paste0(BASE, "Data_Group_4.xlsx"), sheet = "Portfolio")
```

```
lapply(list(ratings, portfolio), setDT)
```

```
## [[1]]
##      Obligor 2024.12.31 2025.01.31 2025.02.28 2025.03.31 2025.04.30
##      <num>      <char>      <char>      <char>      <char>      <char>
## 1:         1      CCC/C      CCC/C      CCC/C          B          B
## 2:         2      BBB      BBB      BBB      BBB      BBB
## 3:         3      BB       BB       BB       BB       BB
## 4:         4      B       B       B       B       B
## 5:         5      AAA      AAA      AAA      AAA      AAA
## ---
## 9996:      9996      BB       BB       A       A       A
## 9997:      9997      BB       BB       BB      BB      BB
## 9998:      9998      CCC/C      CCC/C      CCC/C      CCC/C      CCC/C
## 9999:      9999      BB       BB       BB      BB      BB
## 10000:     10000      AA       AA       AA      AA      AA
##      2025.05.31 2025.06.30 2025.07.31 2025.08.31 2025.09.30 2025.10.31
##      <char>      <char>      <char>      <char>      <char>      <char>
## 1:         B      B      B      B      B      B
## 2:      BBB      BBB      BBB      BBB      BBB      BBB
## 3:         BB      BB      BB      BB      BB      BB
## 4:         B      B      B      B      B      B
## 5:      AAA      AAA      AAA      AAA      AAA      AAA
## ---
## 9996:         A      A      A      A      A      A
## 9997:         BB      BB      BB      BB      BB      BB
## 9998:      CCC/C      CCC/C      CCC/C      CCC/C      CCC/C      CCC/C
## 9999:         BB      BB      BB      BB      BB      BB
## 10000:         AA      AA      AA      AA      AA      AA
##      2025.11.30 2025.12.31
##      <char>      <char>
## 1:         B      B
## 2:      BBB      BBB
## 3:         BB      BB
## 4:         B      B
## 5:      AAA      AAA
## ---
## 9996:         A      A
## 9997:         BB      BB
## 9998:      CCC/C      CCC/C
## 9999:         BB      BB
## 10000:         AA      AA
##
```

[[2]]

##	Obligor	Rating	Industry	Loan.Volume	
##	<num>	<char>	<num>	<num>	
##	1:	1	BBB	1	169000
##	2:	2	BB	3	149000
##	3:	3	BB	2	850000
##	4:	4	BB	2	76000
##	5:	5	CCC/C	1	174000
##	6:	6	BBB	1	108000
##	7:	7	A	1	122000
##	8:	8	A	3	33000
##	9:	9	BB	3	22000
##	10:	10	BBB	3	38000
##	11:	11	AAA	1	142000
##	12:	12	BB	2	113000
##	13:	13	A	3	850000
##	14:	14	B	3	33000
##	15:	15	BB	2	166000
##	16:	16	BB	1	84000
##	17:	17	BBB	1	143000
##	18:	18	A	1	191000
##	19:	19	B	2	96000
##	20:	20	BB	3	90000
##	21:	21	BB	3	850000
##	22:	22	BBB	1	850000
##	23:	23	A	1	850000
##	24:	24	A	1	198000
##	25:	25	BB	3	162000
##	26:	26	BB	2	21000
##	27:	27	CCC/C	1	850000
##	28:	28	BBB	2	108000
##	29:	29	BBB	2	49000
##	30:	30	BB	3	50000
##	31:	31	BBB	1	48000
##	32:	32	A	1	33000
##	33:	33	BB	1	137000
##	34:	34	AAA	1	72000
##	35:	35	AA	3	103000
##	36:	36	AA	2	91000
##	37:	37	BBB	1	182000
##	38:	38	A	2	88000
##	39:	39	BBB	2	44000
##	40:	40	B	3	37000
##	41:	41	BB	3	52000
##	42:	42	BB	3	79000
##	43:	43	A	2	46000
##	44:	44	BB	3	144000
##	45:	45	A	3	94000

##	46:	46	A	1	143000
##	47:	47	AA	1	850000
##	48:	48	CCC/C	1	21000
##	49:	49	BBB	2	168000
##	50:	50	BBB	1	148000
##	51:	51	BBB	2	193000
##	52:	52	BBB	3	193000
##	53:	53	A	2	119000
##	54:	54	BBB	1	85000
##	55:	55	BBB	1	139000
##	56:	56	BBB	3	178000
##	57:	57	AA	2	21000
##	58:	58	BB	2	172000
##	59:	59	AA	1	31000
##	60:	60	CCC/C	1	52000
##	61:	61	A	2	125000
##	62:	62	BBB	1	34000
##	63:	63	AA	1	59000
##	64:	64	BB	2	44000
##	65:	65	AAA	1	65000
##	66:	66	BBB	2	91000
##	67:	67	BBB	1	850000
##	68:	68	BBB	1	56000
##	69:	69	B	2	49000
##	70:	70	A	1	57000
##	71:	71	BBB	3	97000
##	72:	72	BB	2	64000
##	73:	73	BBB	1	74000
##	74:	74	BB	3	149000
##	75:	75	BB	2	24000
##	76:	76	B	2	121000
##	77:	77	A	1	198000
##	78:	78	A	3	190000
##	79:	79	A	3	93000
##	80:	80	BBB	3	110000
##	81:	81	B	2	137000
##	82:	82	BB	2	151000
##	83:	83	CCC/C	1	111000
##	84:	84	BBB	3	94000
##	85:	85	BB	3	200000
##	86:	86	CCC/C	1	113000
##	87:	87	BB	1	135000
##	88:	88	AA	1	60000
##	89:	89	BBB	2	103000
##	90:	90	B	1	67000
##	91:	91	BBB	3	166000
##	92:	92	AA	2	195000
##	93:	93	B	1	184000

```
## 94:      94      BB      1      186000
## 95:      95     BBB      1      139000
## 96:      96       B      3      163000
## 97:      97     BBB      1       21000
## 98:      98      BB      3      138000
## 99:      99      BB      3      120000
## 100:     100     BBB      2      143000
##      Obligor Rating Industry Loan.Volume
##      <num> <char>      <num>      <num>
```

```
lapply(list(ratings, portfolio), str)
```

```
## Classes 'data.table' and 'data.frame':  10000 obs. of  14 variables:
## $ Obligor   : num  1 2 3 4 5 6 7 8 9 10 ...
## $ 2024.12.31: chr  "CCC/C" "BBB" "BB" "B" ...
## $ 2025.01.31: chr  "CCC/C" "BBB" "BB" "B" ...
## $ 2025.02.28: chr  "CCC/C" "BBB" "BB" "B" ...
## $ 2025.03.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.04.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.05.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.06.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.07.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.08.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.09.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.10.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.11.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.12.31: chr  "B" "BBB" "BB" "B" ...
## Classes 'data.table' and 'data.frame':  100 obs. of  4 variables:
## $ Obligor   : num  1 2 3 4 5 6 7 8 9 10 ...
## $ Rating     : chr  "BBB" "BB" "BB" "BB" ...
## $ Industry   : num  1 3 2 2 1 1 1 3 3 3 ...
## $ Loan.Volume: num  169000 149000 850000 76000 174000 108000 122000 33000 22000 38000 ...

## [[1]]
## NULL
##
## [[2]]
## NULL
```

```
lapply(list(ratings, portfolio), head)
```

```
## [[1]]
##      Obligor 2024.12.31 2025.01.31 2025.02.28 2025.03.31 2025.04.30 2025.05.31
##      <num>      <char>      <char>      <char>      <char>      <char>      <char>
## 1:         1      CCC/C      CCC/C      CCC/C          B          B          B
## 2:         2       BBB       BBB       BBB       BBB       BBB       BBB
## 3:         3        BB        BB        BB        BB        BB        BB
## 4:         4         B         B         B         B         B         B
## 5:         5       AAA       AAA       AAA       AAA       AAA       AAA
## 6:         6       BBB       BBB       BBB       BBB       BBB       BBB
```

```
##      2025.06.30 2025.07.31 2025.08.31 2025.09.30 2025.10.31 2025.11.30 2025.12.31
##      <char>      <char>      <char>      <char>      <char>      <char>      <char>
## 1:      B        B        B        B        B        B        B
## 2:      BBB      BBB      BBB      BBB      BBB      BBB      BBB
## 3:      BB       BB       BB       BB       BB       BB       BB
## 4:      B        B        B        B        B        B        B
## 5:      AAA      AAA      AAA      AAA      AAA      AAA      AAA
## 6:      BBB      BBB      BBB      BBB      BBB      BBB      BBB
```

```
##
```

```
## [[2]]
```

```
##      Obligor Rating Industry Loan.Volume
##      <num> <char>      <num>      <num>
## 1:      1      BBB      1        169000
## 2:      2      BB       3        149000
## 3:      3      BB       2        850000
## 4:      4      BB       2        76000
## 5:      5      CCC/C    1        174000
## 6:      6      BBB      1        108000
```

```
lapply(list(ratings, portfolio), summary)
```

```
## [[1]]
```

```
##      Obligor      2024.12.31      2025.01.31      2025.02.28
## Min.      : 1      Length :10000      Length :10000      Length :10000
## 1st Qu.: 2501      N.unique : 7      N.unique : 8      N.unique : 8
## Median : 5000      N.blank : 0      N.blank : 0      N.blank : 0
## Mean    : 5000      Min.nchar: 1      Min.nchar: 1      Min.nchar: 1
## 3rd Qu.: 7500      Max.nchar: 5      Max.nchar: 5      Max.nchar: 5
## Max.      :10000
##      2025.03.31      2025.04.30      2025.05.31      2025.06.30
## Length :10000      Length :10000      Length :10000      Length :10000
## N.unique : 8      N.unique : 8      N.unique : 8      N.unique : 8
## N.blank : 0      N.blank : 0      N.blank : 0      N.blank : 0
## Min.nchar: 1      Min.nchar: 1      Min.nchar: 1      Min.nchar: 1
## Max.nchar: 5      Max.nchar: 5      Max.nchar: 5      Max.nchar: 5
##
##      2025.07.31      2025.08.31      2025.09.30      2025.10.31
## Length :10000      Length :10000      Length :10000      Length :10000
## N.unique : 8      N.unique : 8      N.unique : 8      N.unique : 8
## N.blank : 0      N.blank : 0      N.blank : 0      N.blank : 0
## Min.nchar: 1      Min.nchar: 1      Min.nchar: 1      Min.nchar: 1
## Max.nchar: 5      Max.nchar: 5      Max.nchar: 5      Max.nchar: 5
##
##      2025.11.30      2025.12.31
## Length :10000      Length :10000
## N.unique : 8      N.unique : 8
## N.blank : 0      N.blank : 0
## Min.nchar: 1      Min.nchar: 1
```

```
## Max.nchar:    5    Max.nchar:    5
##
##
## [[2]]
##      Obligor          Rating      Industry      Loan.Volume
## Min.   : 1.00    Length   :100    Min.   :1.00    Min.   : 21000
## 1st Qu.: 25.75    N.unique  : 7    1st Qu.:1.00    1st Qu.: 59750
## Median : 50.50    N.blank  : 0    Median :2.00    Median :112000
## Mean   : 50.50    Min.nchar: 1    Mean   :1.87    Mean   :165760
## 3rd Qu.: 75.25    Max.nchar: 5    3rd Qu.:3.00    3rd Qu.:166000
## Max.   :100.00                Max.   :3.00    Max.   :850000

nrow(ratings)

## [1] 10000

nrow(portfolio)

## [1] 100

print(colnames(ratings))

## [1] "Obligor"      "2024.12.31" "2025.01.31" "2025.02.28" "2025.03.31"
## [6] "2025.04.30" "2025.05.31" "2025.06.30" "2025.07.31" "2025.08.31"
## [11] "2025.09.30" "2025.10.31" "2025.11.30" "2025.12.31"

print(colnames(portfolio))

## [1] "Obligor"      "Rating"      "Industry"      "Loan.Volume"
# Correctly parse the dots and format to dashes
colnames(ratings)[-1] <- format(as.Date(colnames(ratings)[-1]), format = "%Y.%m.%d"), "%Y-%m-%d"
print(colnames(ratings))

## [1] "Obligor"      "2024-12-31" "2025-01-31" "2025-02-28" "2025-03-31"
## [6] "2025-04-30" "2025-05-31" "2025-06-30" "2025-07-31" "2025-08-31"
## [11] "2025-09-30" "2025-10-31" "2025-11-30" "2025-12-31"
```

1 Task

1.1

Based on the rating history of 10,000 representative borrowers, determine the one-year migration matrix based on the Lando and Skodeberg procedure. In doing so, determine the migration matrix using the generator matrix based on at least the first five terms.

Estimation of the generator matrix come from this formula:

$$\lambda_{ij} = \frac{N_{ij}}{\int Y_j(s) ds} \quad i \neq j \quad \text{and} \quad \lambda_{ii} = - \sum_{j \neq i} \lambda_{ij}$$

where N_{ij} is the number of observed migrations from rating i to rating j and $Y_j(s)$ is the number of borrowers in rating j at time s . And the integral: $\int Y_j(s) ds$ is the total time spent by all borrowers

in rating j during the observation period. The one-year migration matrix can then be obtained by exponentiating the generator matrix: $W(\Delta t) = e^{\Lambda \cdot \Delta t} = \sum_{k=0}^{\infty} \frac{(\Lambda \cdot \Delta t)^k}{k!}$

```
generator_matrix <- matrix(0, nrow = nrow(ratings) - 1, ncol = nrow(ratings) - 1)
```

2 Task

2.1

Determine the credit risk for the portfolio of 100 borrowers based on the Basel Standardized Approach. Proceed by using the existing ratings as if they were external ratings.